

Similarly, when considering statements about the ceiling function, it is useful to write $x = n + \varepsilon$, where $n = \lceil x \rceil$ is an integer and $0 \leq \varepsilon < 1$.

Example 31: Prove that if x is a real number, then $\lfloor 2x \rfloor = \lfloor x \rfloor + \lfloor x + \frac{1}{2} \rfloor$

Solution: Let, $\lfloor 2x \rfloor = n$. By Defn. 12, $n \in \mathbb{Z}$.

$$\therefore n + \varepsilon_1 = 2x, \text{ where } 0 \leq \varepsilon_1 < 1.$$

Let, $\lfloor x \rfloor = m$. By Defn. 12, $m \in \mathbb{Z}$.

$$\therefore m + \varepsilon_2 = x, \text{ where } 0 \leq \varepsilon_2 < 1.$$

Let, $\lfloor x + \frac{1}{2} \rfloor = p$. By Defn. 12, $p \in \mathbb{Z}$.

$$\therefore p + \varepsilon_3 = x + \frac{1}{2}, \text{ where } 0 \leq \varepsilon_3 < 1.$$

$$(m + \varepsilon_2) + (p + \varepsilon_3) = 2x + \frac{1}{2} \Rightarrow (m + p) + (\varepsilon_2 + \varepsilon_3) = 2x + \frac{1}{2}. \text{ (Wrong Way)}$$

Book's Solution (Difficult)

To prove this statement we let $x = n + \varepsilon$, where n is an integer and $0 \leq \varepsilon < 1$. There are two cases to consider, depending on whether ε is less than, or greater than or equal to $\frac{1}{2}$.

We first consider the case when $0 \leq \varepsilon < \frac{1}{2}$. In this case, $2x = 2n + 2\varepsilon$ and $\lfloor 2x \rfloor = 2n$ because $0 \leq 2\varepsilon < 1$. Similarly, $x + \frac{1}{2} = n + (\frac{1}{2} + \varepsilon)$, so $\lfloor x + \frac{1}{2} \rfloor = n$, because $0 \leq \frac{1}{2} + \varepsilon < 1$. Consequently, $\lfloor 2x \rfloor = 2n$ and $\lfloor x \rfloor + \lfloor x + \frac{1}{2} \rfloor = n + n = 2n$.

Next, we consider the case when $\frac{1}{2} \leq \varepsilon < 1$. In this case, $2x = 2n + 2\varepsilon = (2n+1) + (2\varepsilon - 1)$. Because $0 \leq 2\varepsilon - 1 < 1$, it follows that $\lfloor 2x \rfloor = 2n+1$. Because $\lfloor x + \frac{1}{2} \rfloor = \lfloor n + (\frac{1}{2} + \varepsilon) \rfloor = \lfloor n+1 + (\varepsilon - \frac{1}{2}) \rfloor$ and $0 \leq \varepsilon - \frac{1}{2} < 1$, it follows that $\lfloor x + \frac{1}{2} \rfloor = n+1$. Consequently, $\lfloor 2x \rfloor = 2n+1$ and $\lfloor x \rfloor + \lfloor x + \frac{1}{2} \rfloor = n + (n+1) = 2n+1$. This concludes the proof. $\square$

Example 32: Prove or disprove that $\lceil x+y \rceil = \lceil x \rceil + \lceil y \rceil$ for all real numbers x and y .

Solution: This statement is False. Let $x = y = \frac{1}{2}$. $\lceil x+y \rceil = \lceil \frac{1}{2} + \frac{1}{2} \rceil = \lceil 1 \rceil = 1$
 $\lceil x \rceil + \lceil y \rceil = \lceil \frac{1}{2} \rceil + \lceil \frac{1}{2} \rceil = 1 + 1 = 2 \neq 1$